WATER DISTRIBUTION FAIRNESS ANALYSIS

Algorithm, Flowchart, Pseudocode and Sample Output

1. Algorithm

Start the program.

Display the title: Water Distribution Fairness Analysis.

Store the area names, daily water demands, water supplied to each area over three days, and day names.

Set most_underserved to an empty value and highest_shortage to -1.

For each area:

a. Calculate total supply by adding the supplies for all three days.

b. Calculate total demand = daily demand × number of days.

c. Calculate total shortage = total demand - total supply.

d. Display the area name, total demand, total supply and total shortage.

e. For each day, calculate daily shortage = daily demand - daily supply. If the result is negative, set it to zero.

f. Display the daily supply and daily shortage.

g. If the area's total shortage is greater than highest_shortage, update highest_shortage and most_underserved.

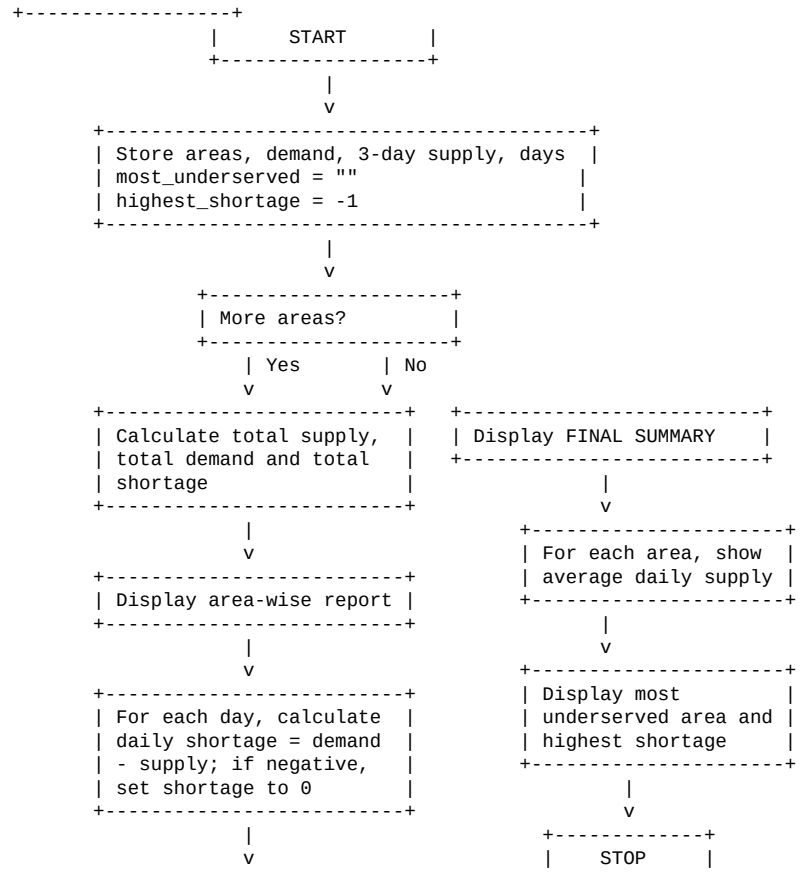
After processing all areas, display the final summary.

For each area, calculate and display average daily supply = total supply ÷ number of days, rounded to two decimal places.

Display the most underserved area and the highest total shortage.

Stop the program.

2. Flowchart



```
       +--------------------------+           +-------------+
       | Is total shortage greater|
       | than highest_shortage?   |
       +--------------------------+
             | Yes         | No
             v             |
       +--------------------------+
       | Update highest_shortage  |
       | and most_underserved     |
       +--------------------------+
             |             |
             +------+------+
                    |
                    +----> Go to next area
```

3. Pseudocode

```
BEGIN
    DISPLAY "WATER DISTRIBUTION FAIRNESS ANALYSIS"

    STORE areas, demand, supply, days
    most_underserved = ""
    highest_shortage = -1

    FOR each area i
        total_supply = SUM(supply[i])
        total_demand = demand[i] * LENGTH(days)
        shortage = total_demand - total_supply

        DISPLAY area, total_demand, total_supply, shortage

        FOR each day j
            daily_shortage = demand[i] - supply[i][j]
            IF daily_shortage < 0 THEN
                daily_shortage = 0
            END IF
            DISPLAY day, supply[i][j], daily_shortage
        END FOR

        IF shortage > highest_shortage THEN
            highest_shortage = shortage
            most_underserved = areas[i]
        END IF
    END FOR

    DISPLAY "FINAL SUMMARY"

    FOR each area i
        average_daily_supply = SUM(supply[i]) / LENGTH(days)
        DISPLAY areas[i], ROUND(average_daily_supply, 2)
    END FOR

    DISPLAY most_underserved
    DISPLAY highest_shortage
END
```

4. Sample Output

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WATER DISTRIBUTION FAIRNESS ANALYSIS
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--------- AREA-WISE REPORT ---------

Area: Area A
Total Demand: 3000 litres
Total Supply: 2250 litres
Total Shortage: 750 litres
Day 1 - Supply: 800 litres, Shortage: 200 litres
Day 2 - Supply: 700 litres, Shortage: 300 litres
Day 3 - Supply: 750 litres, Shortage: 250 litres

Area: Area B
Total Demand: 3600 litres
Total Supply: 3000 litres
Total Shortage: 600 litres
Day 1 - Supply: 1000 litres, Shortage: 200 litres
Day 2 - Supply: 1100 litres, Shortage: 100 litres
Day 3 - Supply: 900 litres, Shortage: 300 litres

Area: Area C
Total Demand: 2400 litres
Total Supply: 2350 litres
Total Shortage: 50 litres
Day 1 - Supply: 800 litres, Shortage: 0 litres
Day 2 - Supply: 750 litres, Shortage: 50 litres
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Day 3 - Supply: 800 litres, Shortage: 0 litres

Area: Area D
Total Demand: 4500 litres
Total Supply: 2550 litres
Total Shortage: 1950 litres
Day 1 - Supply: 900 litres, Shortage: 600 litres
Day 2 - Supply: 800 litres, Shortage: 700 litres
Day 3 - Supply: 850 litres, Shortage: 650 litres

--------- FINAL SUMMARY ---------
Area A - Average Daily Supply: 750.0 litres
Area B - Average Daily Supply: 1000.0 litres
Area C - Average Daily Supply: 783.33 litres
Area D - Average Daily Supply: 850.0 litres

Most Underserved Area: Area D
Highest Total Shortage: 1950 litres
```

Note: This sample output uses the area names, demands and three-day supply values in the supplied Python program. Daily shortages below zero are reported as zero, as in the program.